

precog-esgf-intake: Automated discovery, validation, and optimized download management of Earth system model data from Earth System Grid Federation nodes.

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Summary

precog-esgf-intake is a Python command-line-interface (CLI) software for automated discovery, checking, validation, and download of Earth System Model outputs from Earth System Grid Federation (ESGF) archives. The software is designed for research workflows that require reproducible access to large, distributed climate-model datasets and is particularly aimed at bulk screening of Earth System archives before downstream analysis. The software builds on inherited ESGF access functionality from intake-esgf (Collier et al., 2026), while introducing interactive CLI workflows for archive interrogation, file-availability checks, grid-consistency cross-checks, temporal validation, export of search results in tabular formats, and optimised download management of shortlisted Earth System Model data products.

Statement of need

Modern Earth System science workflows frequently rely on climate-model data distributed across Earth System Grid Federation (ESGF) nodes (Cimadevilla et al., 2025). For example, a Coupled Model Intercomparison Project (CMIP) analysis to constrain future projections in ocean carbon inventories requires standardized output of ocean biogeochemical variables (carbon, oxygen, nutrients) across many models and experiments (Wilson et al., 2022). These archived data are often distributed across many federated storage nodes, including duplicated, incomplete or corrupted versions. Although ESGF provides a federated infrastructure for searching and accessing these archives, practical research workflows often require additional automation to determine whether a given combination of variables, experiments, ensemble members, and grid configurations is both scientifically suitable and operationally downloadable before substantial time is spent retrieving files.

This issue is especially important for CMIP-style analyses in which researchers may need to confirm that the same Earth System Model provides paired pre-industrial, historical and ssp-scenario simulations, that multiple variables of interest are available and whether these are separately archived on compatible grids for downstream analyses; and that temporal coverage is continuous across archived files. Manual inspection of catalogue search results can become slow, repetitive, and error-prone when screening many candidate models or variables across multiple ESGF nodes.

Among its features, precog-esgf-intake provides an interactive workflow for shortlisting Earth System Model datasets that satisfy compound criteria across experiments and variables. The software validates temporal coverage, grid consistency, and file availability across published ESGF archives before download. Users can export search results as tabular summaries, inspect and refine shortlisted datasets, and initiate batch downloads or trigger file integrity checks on

41 locally assigned output paths. Retrieved data are then organised into a directory structure
42 suitable for reproducible downstream analysis. This workflow is particularly useful when
43 researchers need to screen many candidate models and variables before selecting datasets that
44 are both scientifically appropriate and operationally accessible within their computational and
45 storage constraints.

46 Furthermore, by automating the retrieval and validation of CMIP grid metrics like volcello
47 , areacello as default (and enabling further bespoke searches for deptho or thkcello as
48 a validated substitutes when volcello is unavailable) — precog-esgf-intake reduces the
49 need for users to understand CMIP metadata and ESGF directory conventions. By wrapping
50 ESGF discovery in a simple Python CLI and allowing users to tailor searches by editing a
51 straightforward search_criteria.toml configuration file, the package supports both accessible
52 default workflows and more bespoke dataset selection. This lowers the barrier for researchers
53 new to CMIP while enabling experienced users to specify models, experiments, variables,
54 frequencies, and other search constraints. Its lightweight design also supports terminal-based
55 HPC workflows, allowing data discovery and ingestion from visualisation nodes and Jupyter
56 sessions rather than interactive web portals or manual downloads.

57 State of the field

58 CMIP data are currently accessible through the ESGF metagrid web application, which provides
59 web-based discovery and download services for CMIP5, CMIP6, CMIP6Plus, and the upcoming
60 CMIP7 phases. Recent development efforts have focused on analysis-ready, cloud-optimised
61 approaches based on in-memory object storage (Mizielinski et al., 2026), with catalogue
62 sweep tools enabling more scalable access through Python and xarray, while community
63 evaluation frameworks like ESMValTool(Righi et al., 2020) supporting integrated diagnostics
64 for benchmarking model outputs.

65 precog-esgf-intake builds on the ESGF catalogue node sweeping implementation from
66 intake-esgf (Collier et al., 2026), but targets a different level of the workflow: rather than
67 focusing only on data access primitives and caching data-responses to in-memory use, it
68 provides an interactive command-line environment for ample discovery, screening, validating
69 and managing downloads of published ESGF archive products that satisfy scientific constraints
70 relevant to Earth system and ocean biogeochemistry applications. The software is intended for
71 situations in which researchers need to move efficiently from broad discovery and catalogue
72 searches to a smaller, analysis-ready subset of model output that satisfies practical and scientific
73 constraints.

74 Key functionality

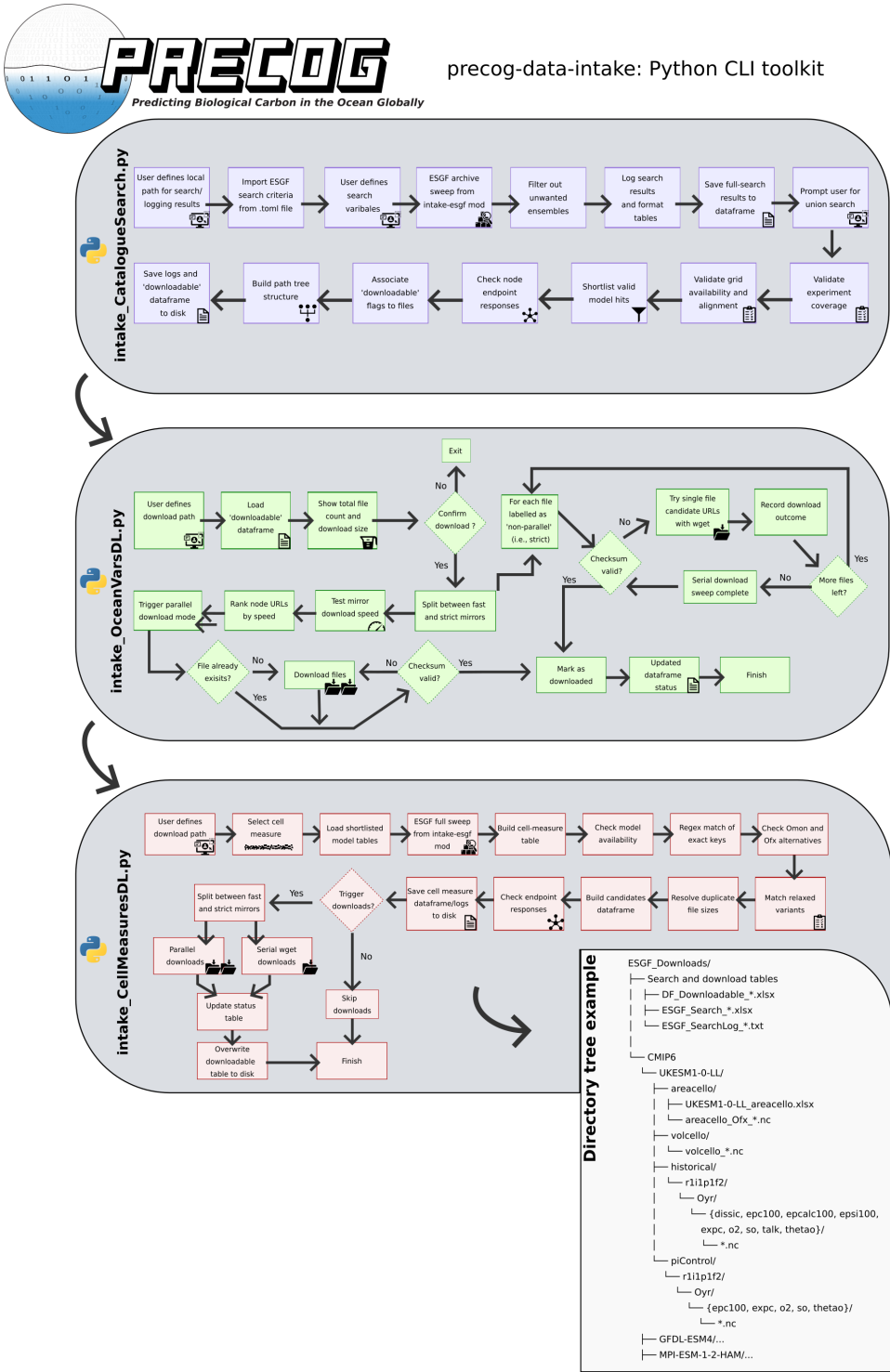
75 The software provides several features tailored to archive-scale Earth system data workflows:

- 76 ■ Definition of ESGF search criteria by modifying a search_criteria.toml configuration
77 file.
- 78 ■ automated ESGF node-based searches using project, activity, experiment, frequency,
79 variable, and grid filters (inherited from intake-esgf);
- 80 ■ interactive user prompts for download paths and variable selections within CLI workflows;
- 81 ■ bulk screening of CMIP6 archive holdings for paired piControl and historical
82 availability;
- 83 ■ validation of continuity in date stamps for CMIP6 pre-industrial and historical simulations;
- 84 ■ checks for consistent availability of model output across compatible grids such as gn and
85 gr;
- 86 ■ conditional multi-variable screening for models that satisfy compound search criteria,
87 including cases where several required variables must be available simultaneously;

- 88 ▪ export of tabular ESGF catalogue search results for easy inspection, search snapshot
- 89 logging, and reproducible shortlist generation;
- 90 ▪ parallel URL pre-checks with verification of server responses and flagging of shortlisted
- 91 outputs as downloadable,
- 92 ▪ download manager with parallel transfers that select responsive ESGF node locations
- 93 and include retry or integrity checks for corrupted files
- 94 ▪ in cases where the same file is available across many nodes, the fastest download option
- 95 is prioritised based on the user's connection
- 96 ▪ retrieval of ocean grid-cell measure variables such as areacello and volcello for selected
- 97 models, together with saved ESGF archive snapshots for later inspection.

98 **Software design and workflow overview**

99 precog-esgf-intake implements a staged workflow for archive-scale ESGF data discovery
100 and retrieval. Rather than moving directly from catalogue search to cached download,
101 the software separates archive interrogation, shortlist generation, downloadability checks,
102 and file retrieval into distinct command-line steps, allowing users to inspect and validate
103 intermediate results before proceeding. In a typical workflow (Figure 1), the user first specifies
104 a parent download directory and one or more target variables. The workflow supports a
105 user-editable `search_criteria.toml` file that can predefine ESGF search facets, including
106 project, experiments, frequency, variables, and grid labels, while still allowing interactive entry
107 of `variable_id` when that field is intentionally left blank. The catalogue search stage then
108 queries ESGF holdings for matching CMIP products (CMIP6 as the default option), filters
109 results to retain scientifically relevant combinations such as paired piControl and historical
110 simulations, and exports tabular search summaries for inspection. Subsequent stages verify
111 whether shortlisted files are reachable on remote nodes, assign local destination paths, and
112 download both target variables and required supporting grid-cell measures such as areacello
113 and volcello. This staged design is intended to improve transparency and reproducibility
114 in archive-based Earth system workflows. By treating search results, validation outputs, and
115 downloadable file lists as explicit intermediate artifacts, the software supports both interactive
116 use and later auditing of dataset selection decisions.



117
118 {width=90%} Figure 1. precog-esgf-intake toolkit overview and directory structure of an
119 example ESGF download. The top-level directory contains search outputs and model-specific
120 CMIP6 data organised by model, experiment, variable, and annual files.

Research impact and applications

A representative use case is the identification of CMIP6 models that simultaneously provide both pre-industrial and historical outputs for ocean biogeochemical variables such as `expc` and `epc100`, together with auxiliary or supporting variables and the associated grid-cell measures required for downstream analyses. The repository accompanying this software includes a [workflow example](#) demonstrating this type of archive screening and retrieval process.

This is particularly relevant for ocean biogeochemistry and carbon-cycle studies, where analyses often depend on coherent combinations of physical and biogeochemical fields rather than isolated variables (e.g., retrieval of carbonate system fields as well as ocean state physical variables, (Wilson et al., 2022)). In such cases, the time spent screening archive holdings, checking consistency, and organising downloads can be substantial, and purpose-built automation improves both efficiency and reproducibility. This data discovery and retrieval functionality by `precog-esgf-intake` is also complementary to established CMIP preprocessing packages such as `xMIP` (Busecke et al., 2023) and `ESMValTool` (Righi et al., 2020), which provide tools for harmonising, cataloguing, and preparing model outputs for analysis; together, these packages support a reproducible workflow from archive discovery and dataset selection through to preprocessing and scientific analysis.

Therefore, `precog-esgf-intake` fills a workflow gap between catalogue access and scientific analysis. The software capabilities support workflows in which data access is itself a significant part of the scientific process, particularly when analyses depend on assembling coherent, scientifically consistent, and analysis-ready ensemble subsets of Earth system model output from distributed archives.

AI usage disclosure

AI assistance was used solely to improve readability of selected software documentation (Claude Sonnet 5 by Anthropic). No AI tools were used to design, implement, or validate the software. All AI-assisted documentation was reviewed and edited by the authors, who retain full responsibility for its accuracy and content. Authors made all the core design and architectural decisions.

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